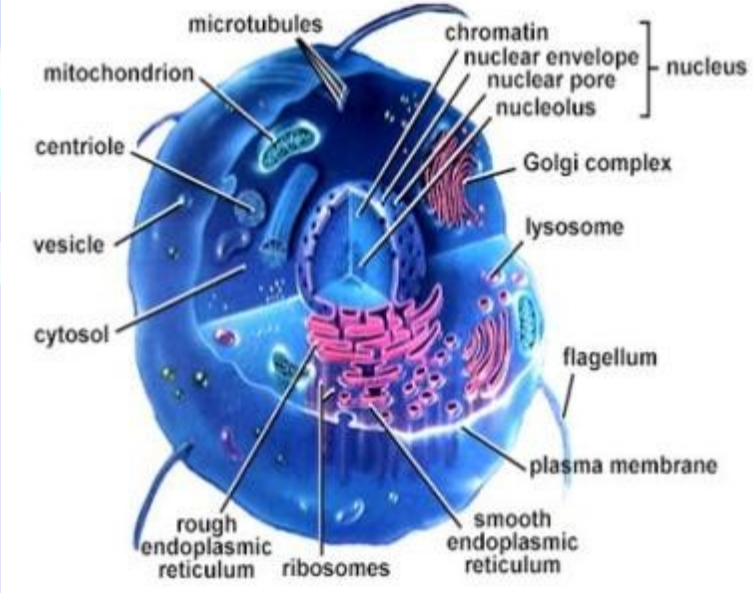
بسم الله الرحمن الرحيم

NUCLEUS



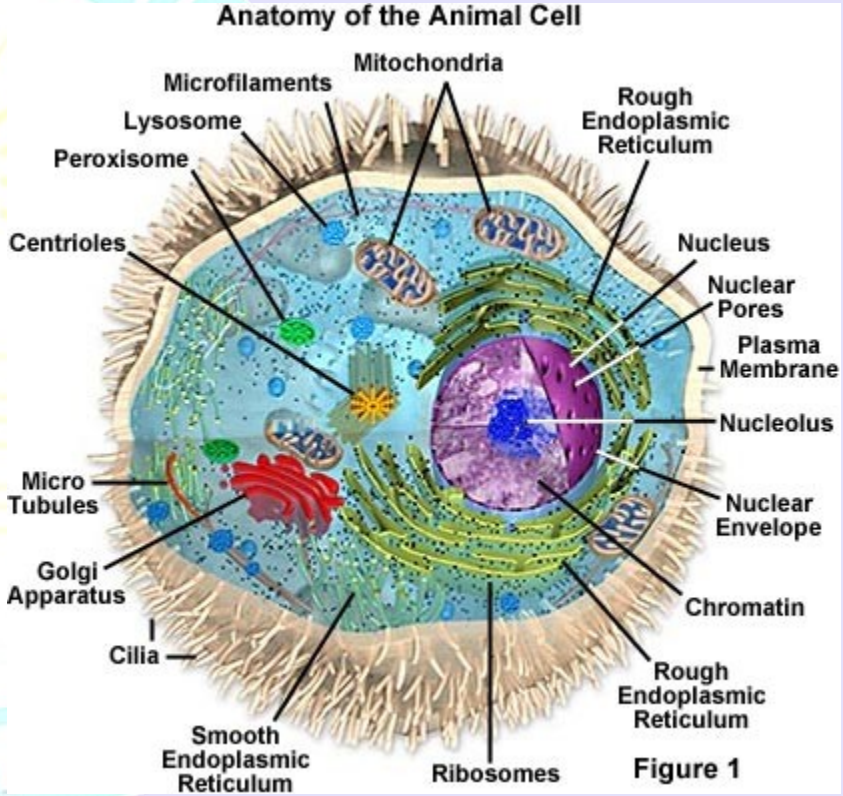
LEARNING OBJECTIVES

At the end of the lecture, students should be able to:

- Define nucleus
- Describe the structure of eukaryotic nucleus
- Identify the important features and functions of nuclear membrane, nucleolus, chromatin and nucleoplasm.

INTRODUCTION (NUCLEUS)

- “Nucleus” is a Latin word meaning **Kernel**.
- It is the “**CONTROL CENTER**” of the cell
- It is **membrane bound** organelle found in **eukaryotic cells**
- It contains the genome, the cell’s database, which is encoded in molecules of the nucleic acid, DNA



NUCLEUS; STRUCTURE

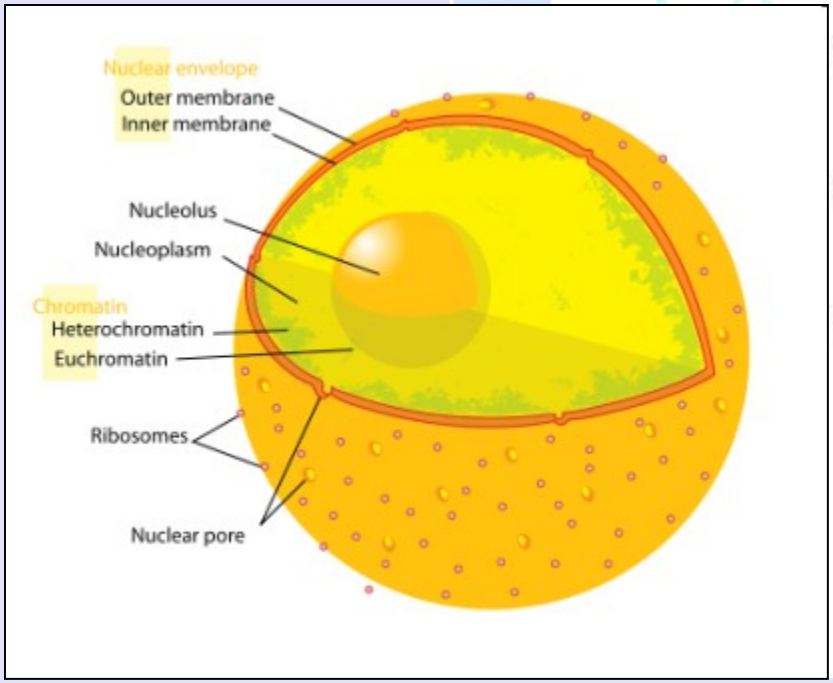
- Consists of

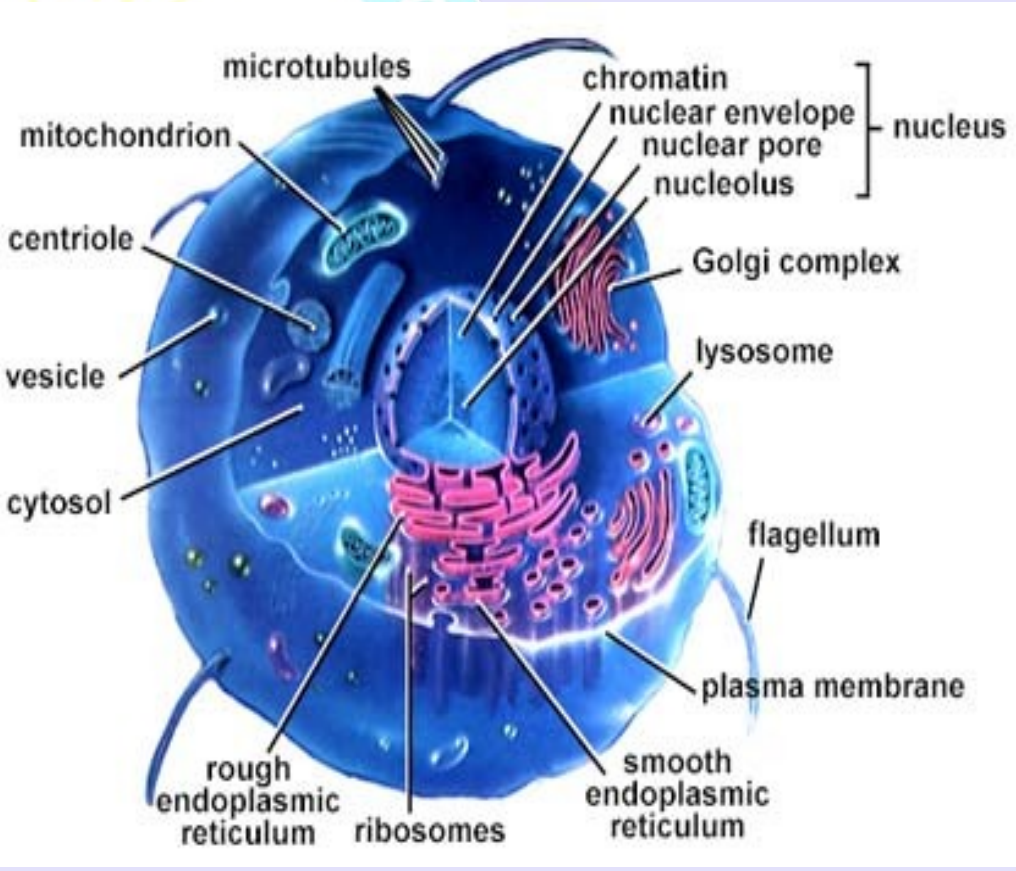
  --Nuclear envelope/membrane

  –Nucleolus

  –Nucleoplasm

  –Chromatin.





NUCLEUS ; FUNCTIONS

This organelle has two major functions:

- stores the cell's **hereditary material,** or DNA.
- **coordinates the cell's activities**, which include growth, intermediary metabolism, protein synthesis, and reproduction (cell division)

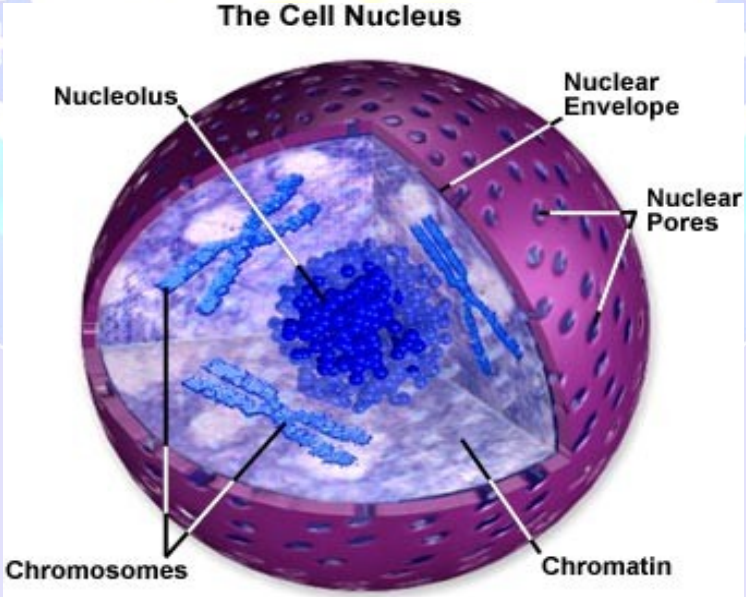


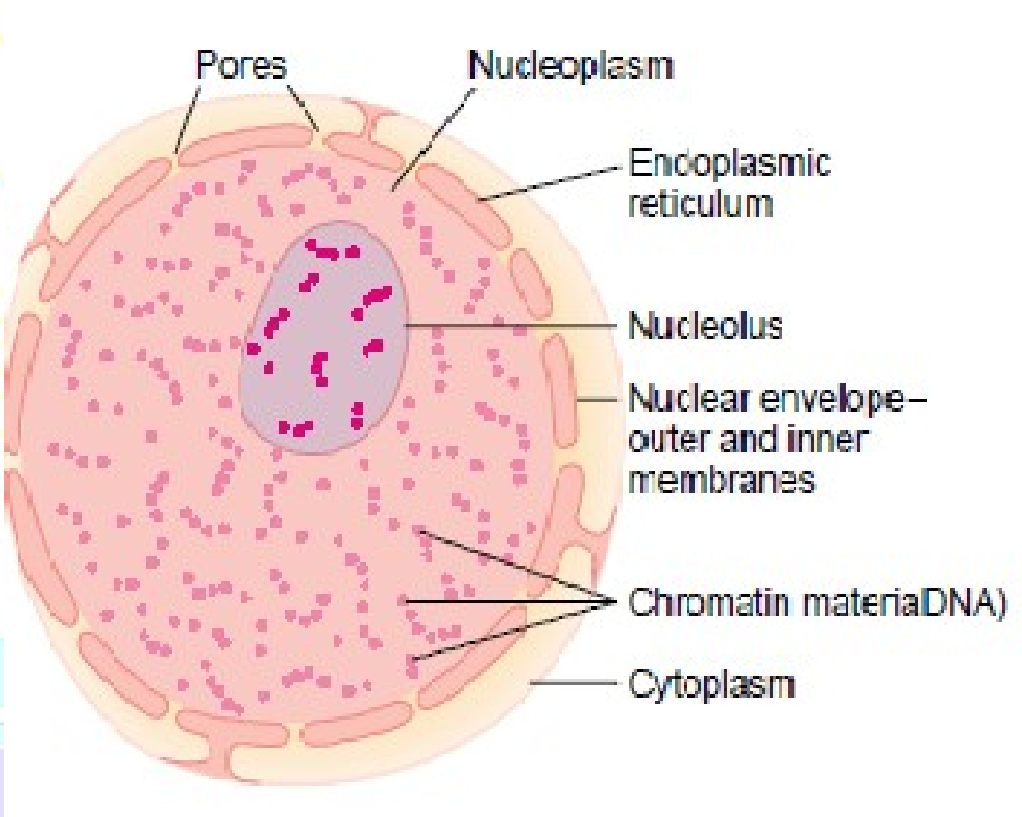
Figure 1

NUCLEAR ENVELOPE/MEMBRANE

- Also known as **perinuclear envelope**.Surrounds the nuclear material.
- It is a lipid bilayer

• Consists of two parallel membranes, separated from each other by a narrow space (20-40nm wide)

•These membranes fuse at intervals, forming openings in the nuclear envelope called **nuclear pores.**



NUCLEAR MEMBRANE/ENVELOPE

Outer Membrane :

- 6 nm thick Faces cytoplasm and is continuous at certain sites with RER
- Ribosomes stud the cytoplasmic surface of the outer nuclear membrane
- Ribosomes synthesize proteins that enter the perinuclear space .

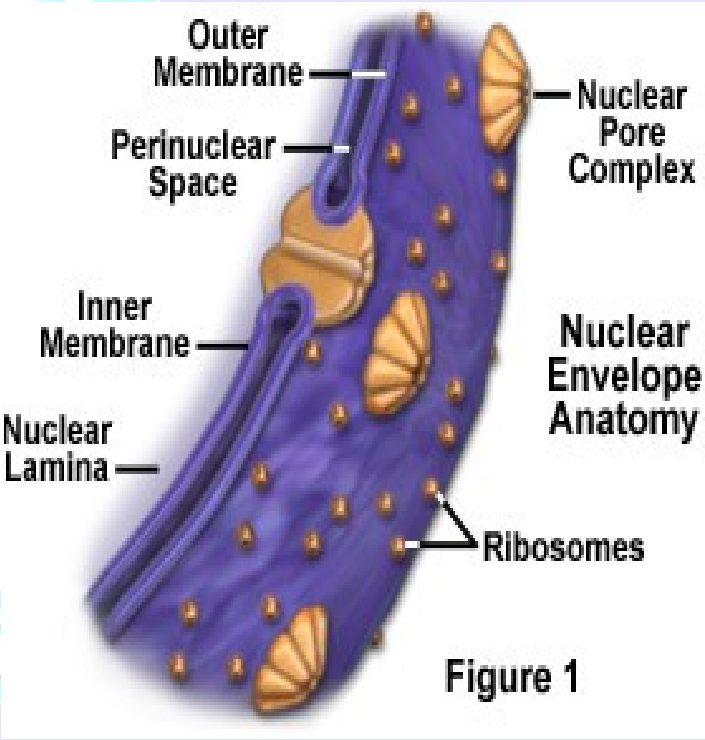


Figure 1

NUCLEAR MEMBRANE/ENVELOPE

Inner membrane :

- Faces the nuclear material
- Outer & inner membranes are fused at nuclear pore complex sites
- Inner Membrane is supported underneath by Nuclear lamina.

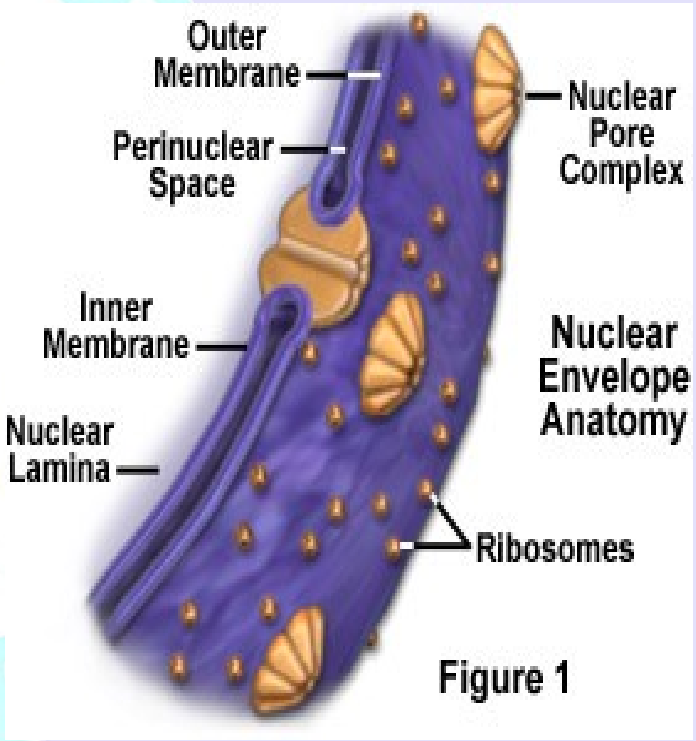
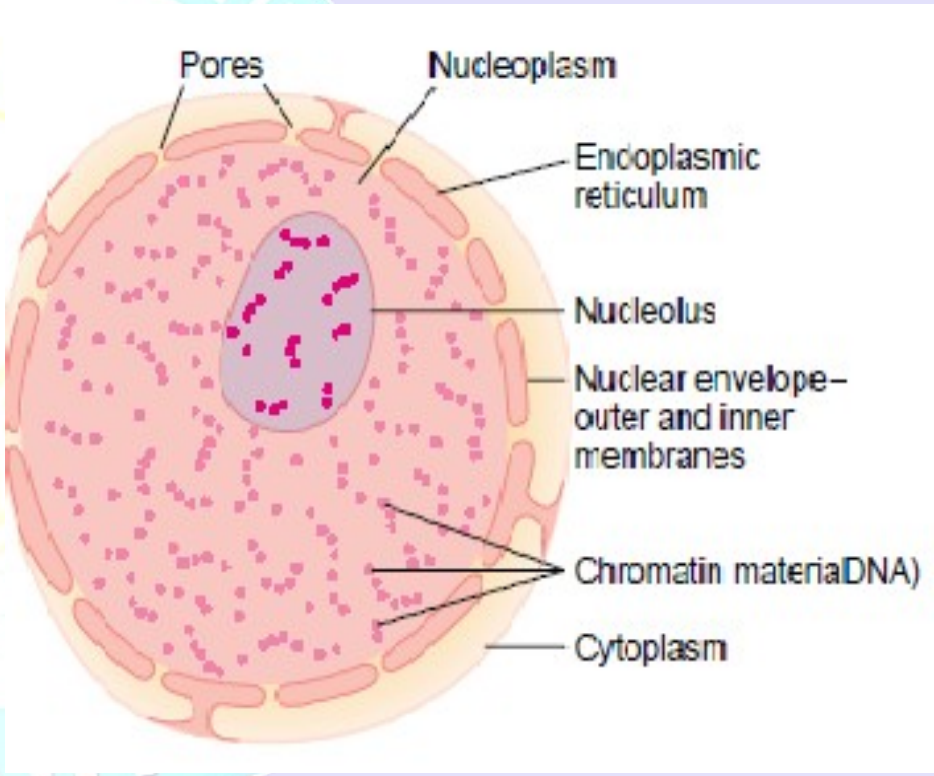


Figure 1

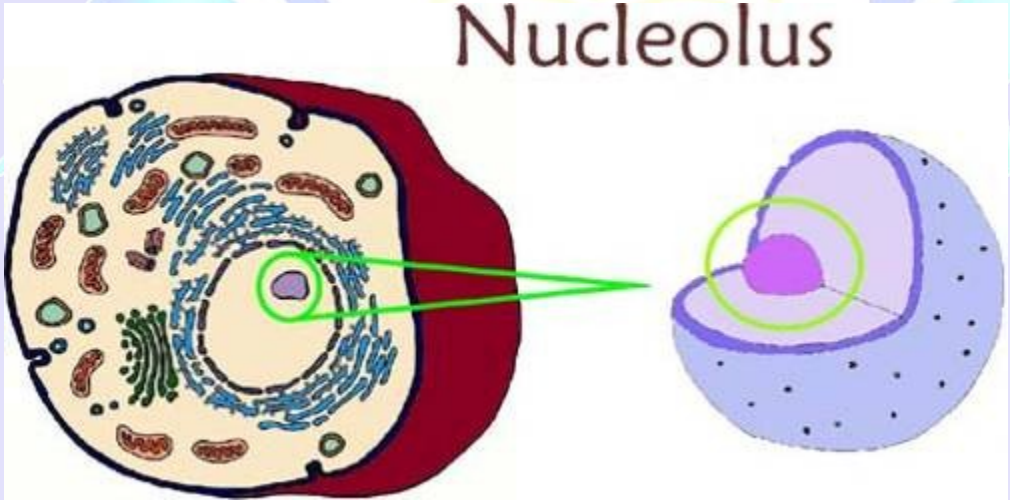
NUCLEAR PORES AND COMPLEX

- Nuclear pores are aqueous channels through the nuclear envelope
- Composed of multiple proteins called nucleoporins
- Nucleus of a cell has 3000 - 4000 pores
- Pores are formed by fusion of outer and inner nuclear membranes
- **Permits passive movement across the nuclear envelope** via 9 -11nm channels by simple diffusion



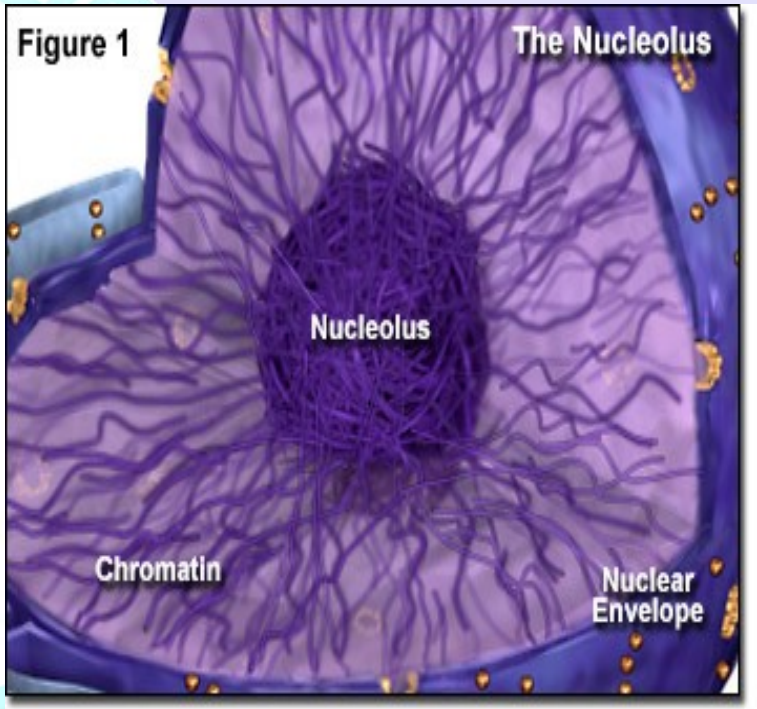
NUCLEOLUS/NUCLEOLI

- The nuclei of most cells contain one or more highly staining structures called *nucleoli.*
- *The nucleolus,* unlike most other organelles, **does not have a limiting membrane.**
- Instead, it is simply an accumulation of **large amounts of RNA and proteins of the types found in ribosomes.**
- The nucleolus becomes enlarged when the cell is actively synthesizing proteins.



NUCLEOLUS/NUCLEOLI

- The nucleolus is comprised of granular and fibrillar components, as well as an ill-defined matrix, in addition to DNA.
- The granular material consists of ribosomal subunits that have already been formed but have not yet matured and are waiting to be exported to the cytoplasm.
- The threadlike fibrillar part of a nucleolus is predominantly composed of rRNA molecules and associated proteins that have joined together to form fibrils.

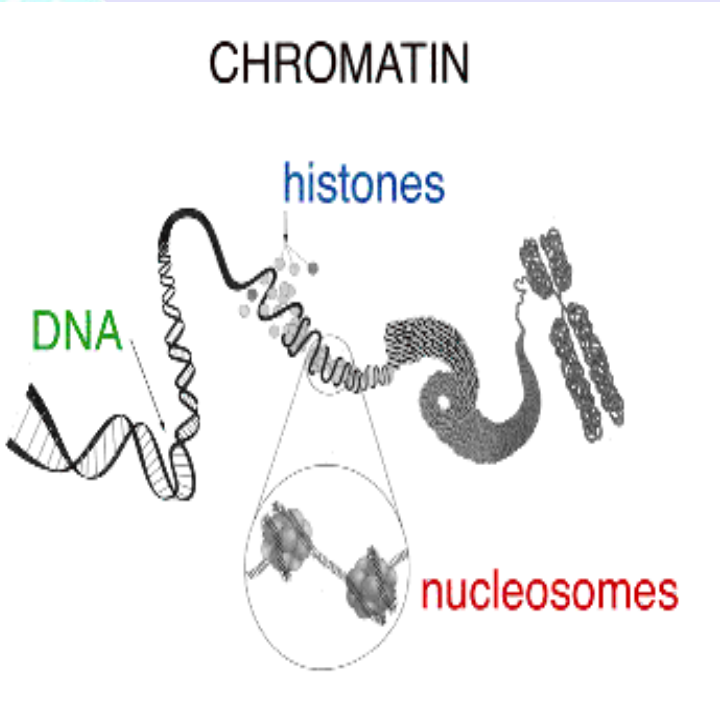


NUCLEOPLASM

- Nucleoplasm is the **protoplasm within the nuclear envelope**
- It consists of a nuclear matrix and various types of particles
- Highly viscous liquid which scaffolds **chromosomes, nucleolus and various granules** like heterochromatin, perichromatin granules
- Many substances like nucleotides and certain enzymes are also dissolved in it.

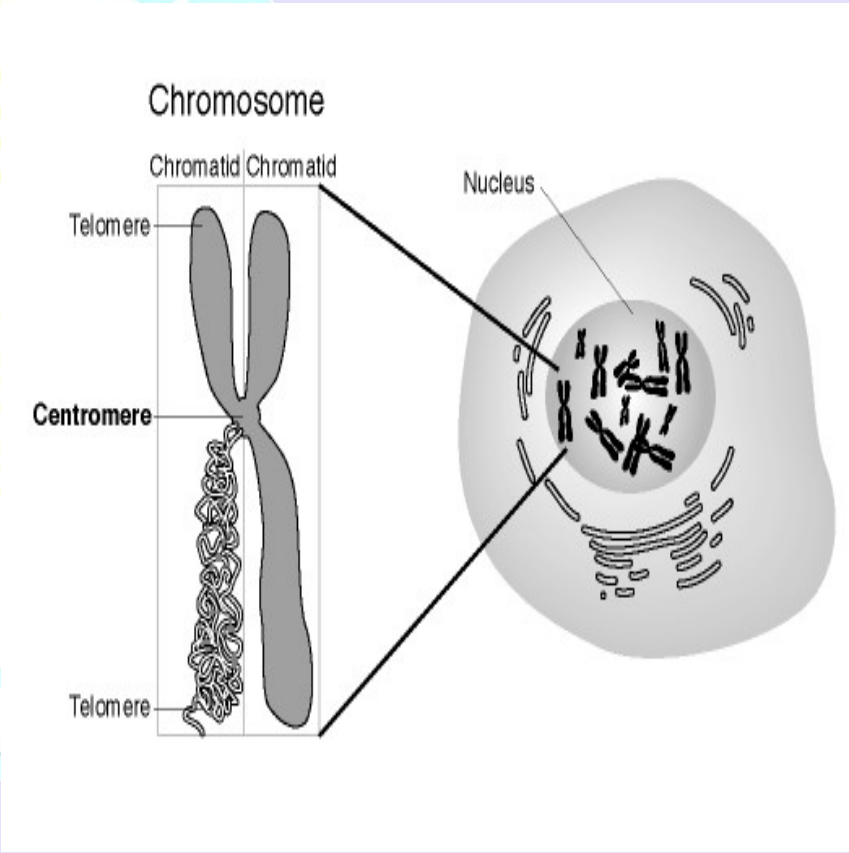
CHROMATIN

- **Chromatin** is the combination of DNA, histone, and other proteins that makes up chromosomes.
- It is found inside the nuclei of eukaryotic cells.
- There are two types of chromatin.
- **Euchromatin** is the less compact DNA form, and contains genes that are frequently expressed by the cell.
- **Heterochromatin,** is the more compact form, and contains DNA that are infrequently transcribed.
- The structure of chromosomes and chromatin varies throughout the cell cycle



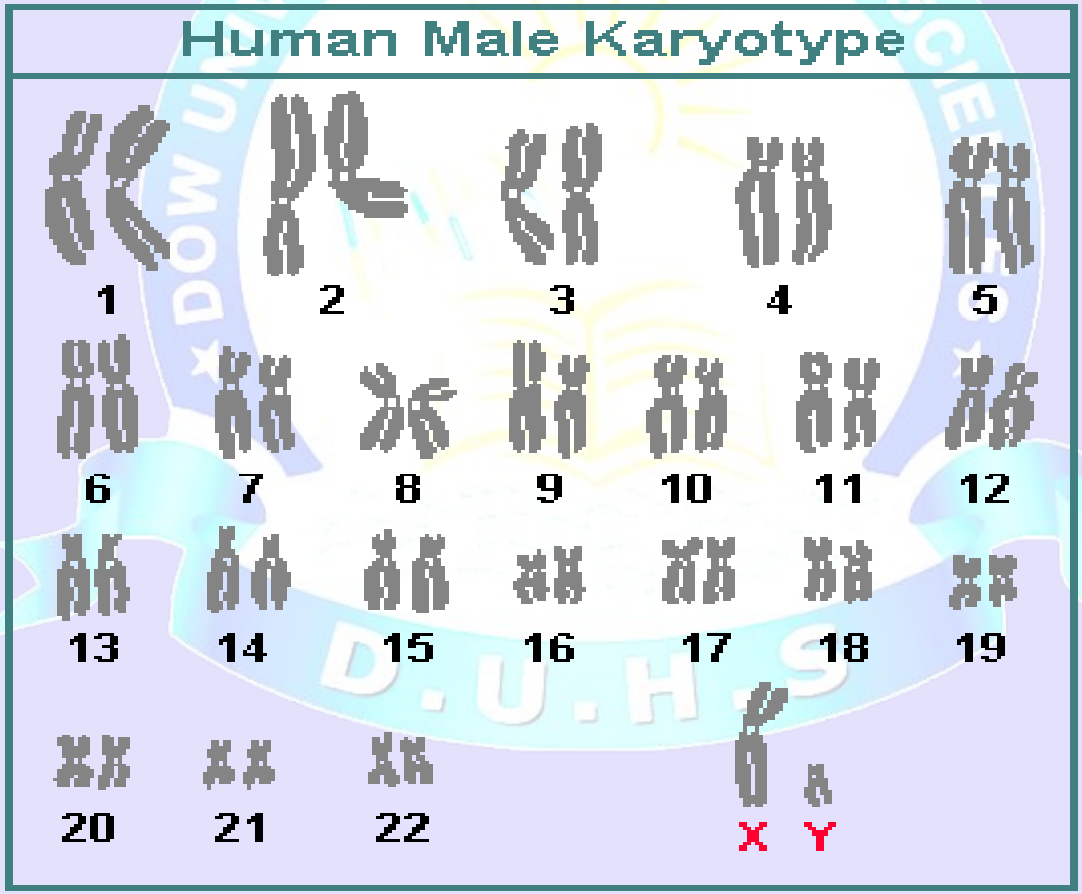
CHROMOSOME

- It is an organized structure of DNA Containing genetic information of cell, regulatory elements and other nucleotide sequences
- Chromosomes may exist as either ;
- **unduplicated** chromosomes are single linear strands, whereas
- **duplicated** chromosomes (copied during synthesis phase) contain two copies joined by a centromere.



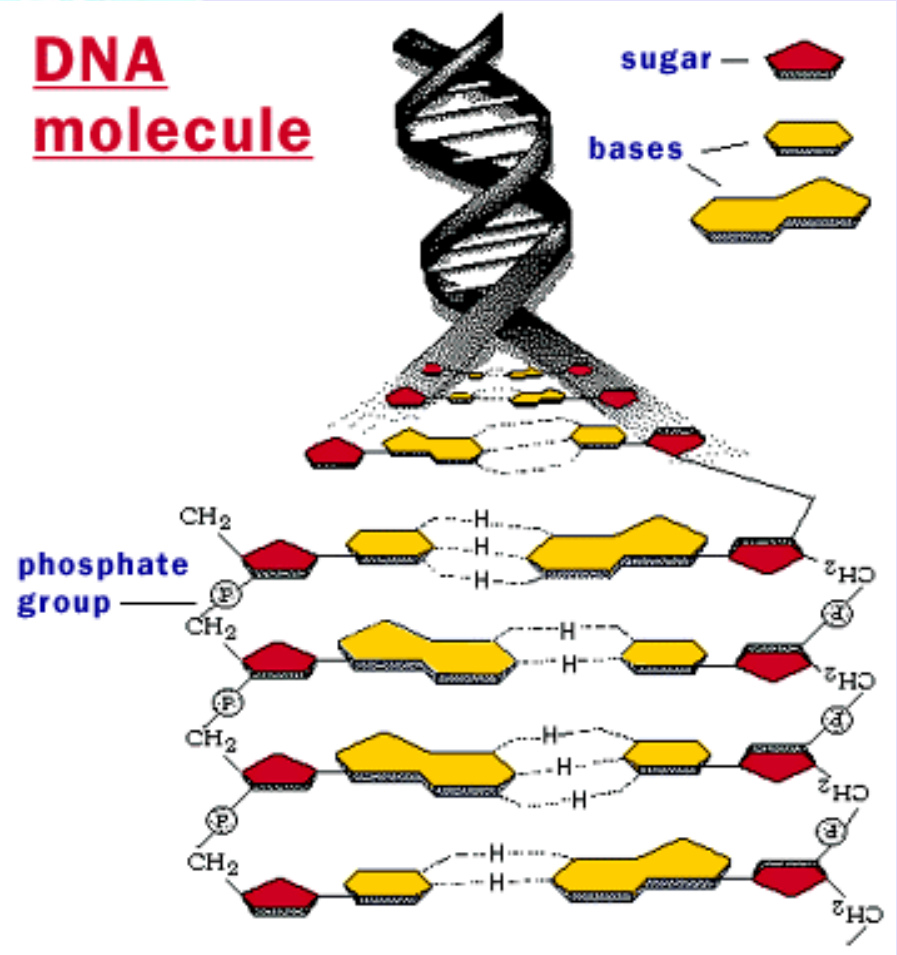
HUMAN CHROMOSOMES

This figure shows the **23 paired chromosomes** of a human male.



DNA MOLECULE

- If the double helix of DNA are set apart, you'll see a complex and sturdy structure.
- Each DNA strand is composed of a **sequence of bases connected on the outside to sugar molecules.**
- These sugars, in turn, are interconnected by phosphate groups.
- On the inside, the bases are connected to bases on the other side of the strand by non-covalent (hydrogen) bonds.



REFERENCES

- **Basic histology by Junqueira Page # 53-59.**


Thank You!